

BAIO LEARNING SYSTEM

State PU Board • CBSE Class 11 • NEET-UG Biology Excellence Series

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SCAN TO CONNECT / BAIO APP

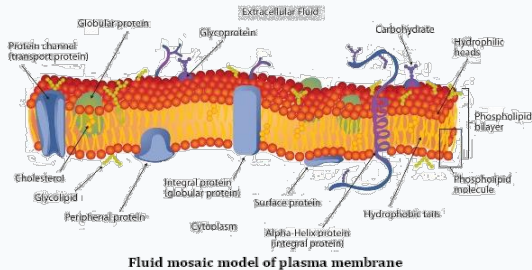
CHAPTER 8: CELL — THE UNIT OF LIFE | BOARD EXAM MANDATORY DIAGRAM GUIDE

PRIORITY-WISE HIGH-YIELD BLUEPRINT

Curated from Board Blueprint Analysis: 5M Long Answers, 3M Explanatory, and 2M Diagrammatic Questions.

★ PRIORITY 1 (MUST LEARN) — FLUID MOSAIC MODEL OF PLASMA MEMBRANE

5 MARKS • SINGER & NICOLSON (1972)



Fluid mosaic model of plasma membrane

Key Structural Concepts:

Continuous phospholipid bilayer in a *quasi-fluid state* allowing lateral movement of proteins ("protein icebergs floating in a sea of lipids"). Amphipathic lipids: polar hydrophilic heads point outward; hydrophobic tails point inward.

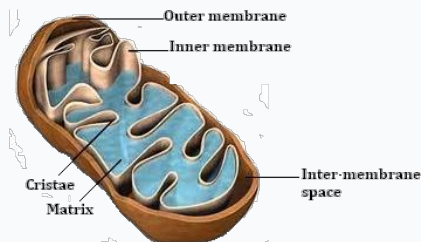
MANDATORY DIAGRAM LABELS (MINIMUM 6 TO 8 FOR FULL MARKS):

Phospholipid bilayer Hydrophilic polar head Hydrophobic fatty acid tail
Integral (Transmembrane) protein Peripheral (Surface) protein Protein channel / pore
Glycoprotein Glycolipid Cholesterol

Board Exam Tip: Draw heads as circles with 2 wavy tails. Clearly distinguish embedded integral proteins from surface peripheral proteins. In human RBCs, membrane is 52% protein, 40% lipid.

★ PRIORITY 1 (MUST LEARN) — ULTRASTRUCTURE OF MITOCHONDRION (L.S.)

3 / 5 MARKS • "POWERHOUSE OF THE CELL"



Structure of mitochondrion (Longitudinal section)

Key Structural Concepts:

Sausage-shaped / cylindrical organelle (0.2–1.0 μm diameter, 1.0–4.1 μm length). Double-membrane envelope dividing the lumen into outer intermembrane space and inner matrix. Inward folds of inner membrane form *cristae* that amplify surface area for ATP synthesis.

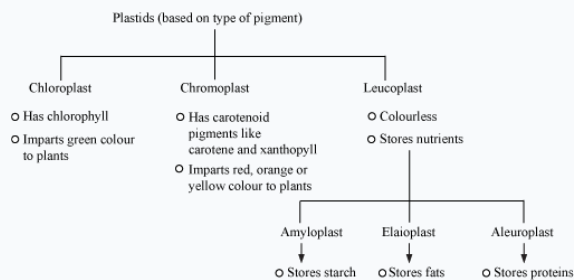
MANDATORY DIAGRAM LABELS:

Outer membrane Inner membrane Intermembrane space Cristae (infoldings)
Mitochondrial matrix Circular dsDNA 70S Ribosomes F_0F_1 particles (Oxysomes)

Board Exam Tip: Draw finger-like cristae projecting into the matrix. Label circular DNA and 70S ribosomes to substantiate its semiautonomous nature. Mention Janus Green B as vital stain.

★ PRIORITY 1 (MUST LEARN) — ULTRASTRUCTURE OF CHLOROPLAST

3 / 5 MARKS • DOUBLE-MEMBRANE SEMIAUTONOMOUS PLASTID



Key Structural Concepts:

Lens-shaped / discoid organelle with a double-membrane envelope. Space inside inner membrane is the *stroma*, containing dark-reaction enzymes, circular DNA, and 70S ribosomes. Flattened sacs called *thylakoids* are stacked like coins into *grana* (light reaction).

MANDATORY DIAGRAM LABELS:

Outer membrane Inner membrane Intermembrane space Stroma (matrix)
Granum (pl. Grana) Thylakoid Thylakoid lumen Stroma lamella (Fret channel)
Chloroplast DNA 70S Ribosomes

Board Exam Tip: Draw grana as coin stacks linked by tubular stroma lamellae. Highlight: Grana = Light Reaction (Photophosphorylation); Stroma = Dark Reaction (Calvin Cycle / CO_2 fixation).

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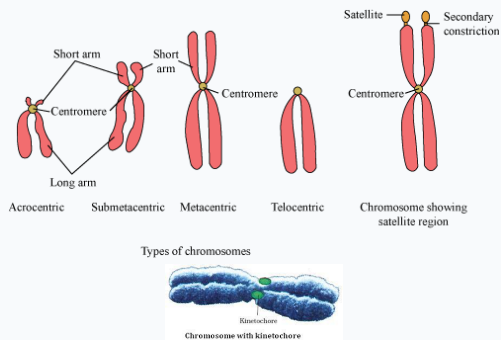
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★ PRIORITY 1 (MUST LEARN) — TYPES OF CHROMOSOMES BASED ON CENTROMERE POSITION

3 / 5 MARKS • MORPHOLOGY & ANAPHASE SHAPES



Morphological Classification & Anaphase Shapes:

- 1. Metacentric:** Centromere strictly in center; two equal arms ($p = q$). **V-shaped** at anaphase.
- 2. Submetacentric:** Centromere slightly displaced; one short p-arm & one long q-arm. **L-shaped** at anaphase.
- 3. Acrocentric:** Centromere situated very close to end; tiny p-arm & very long q-arm. **J-shaped** at anaphase.
- 4. Telocentric:** Centromere strictly terminal; only one arm present. **I-shaped (rod)** at anaphase.

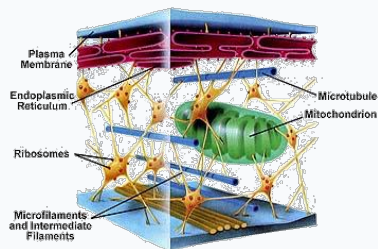
Satellite (SAT) Chromosome: Non-staining secondary constriction distal to centromere.

MANDATORY DIAGRAM LABELS:

Centromere (primary constriction) Kinetochore Short p-arm Long q-arm
 Secondary constriction Satellite region (SAT body)

★ PRIORITY 2 (HIGH PROBABILITY) — ULTRASTRUCTURE OF CILIA / FLAGELLA (9+2 AXONEME T.S.)

3 / 5 MARKS • AXONEMAL MICROTUBULE ARRAY



Key Structural Concepts:

Core structure is the *axoneme*, enclosed by plasma membrane and composed of microtubules running parallel to the axis. Consists of **9 peripheral doublets** arranged radially around **1 central pair of singlets** (the classic **(9 + 2) array**).

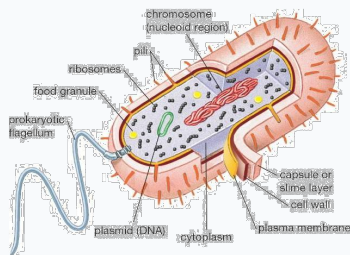
MANDATORY DIAGRAM LABELS:

Plasma membrane Peripheral microtubule doublets (9 pairs)
 Central singlet microtubules (pair) Central sheath Radial spokes (9)
 Interdoublet bridge (Nexin) Dynein motor arms

Board Exam Tip: Draw exactly 9 peripheral doublets and 2 central singlets. Show radial spokes extending inward to the central sheath. Dynein arms provide ATPase motor activity for ciliary beating.

★ PRIORITY 2 (HIGH PROBABILITY) — PROKARYOTIC BACTERIAL CELL STRUCTURE

3 / 5 MARKS • CELL ENVELOPE & CYTOPLASMIC INCLUSIONS



Key Structural Concepts:

Lacks membrane-bound nucleus and organelles. Possesses a tightly bound 3-layered cell envelope: outer glycocalyx (capsule/slime layer), middle rigid peptidoglycan cell wall, and inner semi-permeable plasma membrane with mesosomal infoldings.

MANDATORY DIAGRAM LABELS:

Capsule / Slime layer Cell wall (peptidoglycan) Plasma membrane
 Nucleoid (naked circular DNA) Plasmid (extrachromosomal DNA) 70S Ribosomes
 Mesosome infolding Flagellum Pili & Fimbriae Inclusion bodies

Board Exam Tip: Differentiate flagellum (locomotion: filament, hook, basal body) from pili/fimbriae (attachment to host/substratum). Mesosomes carry respiratory enzymes and aid DNA distribution.

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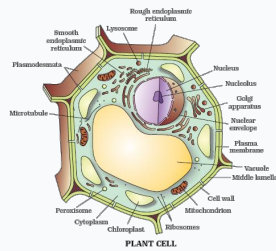


SCAN TO CONNECT / BAIO APP

★ PRIORITY 3 (COMPARATIVE CELL ANATOMY) — EUKARYOTIC PLANT CELL VS. ANIMAL CELL

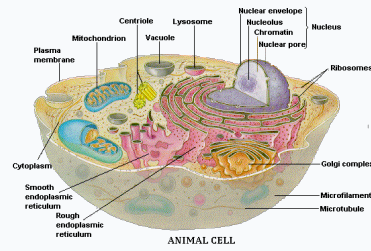
5 MARKS • COMPREHENSIVE ORGANELLE MAP

TYPICAL PLANT CELL



PLANT CELL

TYPICAL ANIMAL CELL



ANIMAL CELL

Key Structures in Plant Cell:

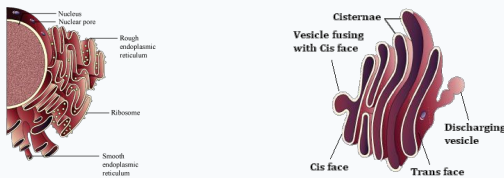
- **Cellulosic Cell Wall:** With middle lamella (calcium pectate).
- **Large Central Vacuole:** Enclosed by tonoplast (90% vol).
- **Plastids:** Chloroplasts, chromoplasts, leucoplasts.
- **Plasmodesmata:** Cytoplasmic bridging between adjacent cells.

Key Structures in Animal Cell:

- **Centrosome & Centrioles:** Cylindrical pair for spindle fibers.
- **Lysosomes:** Abundant digestive suicide bags with acid hydrolases.
- **Microvilli:** Brush border surface infoldings for absorption.
- **Vacuoles:** Small and non-permanent (no central vacuole).

★ PRIORITY 3 (IMPORTANT ORGANELLES) — ENDOPLASMIC RETICULUM & GOLGI APPARATUS

3 MARKS • ENDOMEMBRANE SYSTEM & SECRETION



Endomembrane Functional Coordination:

Rough ER: Studded with 80S ribosomes; primary site for protein synthesis and sorting.

Smooth ER: Tubular network free of ribosomes; site for lipid and steroid synthesis.

Golgi Apparatus: Parallel stacked cisternae (0.5–1.0 µm). Shows polarity: **Cis-face (forming/convex)** receives vesicles → glycosylates to glycoproteins/glycolipids → discharges at **Trans-face (maturing/concave)**.

MANDATORY DIAGRAM LABELS:

Cisternae Cis face Trans face Fusing vesicle Discharging vesicle
Rough ER Smooth ER

★ PRIORITY 4 (SPECIALIZED) — ULTRASTRUCTURE OF NUCLEUS & 70S vs 80S RIBOSOME SUBUNITS

2 / 3 MARKS • NUCLEAR MATRIX & RIBOSOME ARCHITECTURE



Key Structural Details:

Nucleus: Bound by nuclear envelope with perinuclear space (10–50 nm) & nuclear pores. Encloses nucleoplasm, nucleolus (rRNA synthesis site), and chromatin nucleoprotein fibers.

Ribosomes: Prokaryotic 70S = 50S + 30S; Eukaryotic 80S = 60S + 40S (S = Svedberg sedimentation coefficient).

MANDATORY DIAGRAM LABELS:

Nuclear envelope Nuclear pore Nucleoplasm Nucleolus Chromatin
50S & 30S subunits (70S) 60S & 40S subunits (80S)

Examiner's Diagram Scoring Rules (Board Exam Evaluation Scheme):

1. **Neat Outline & Scale (50% Marks):** Draw a clear, proportional pencil outline before adding labels.
2. **Horizontal Guide Lines:** Draw all label lines horizontally to the right with a ruler; do not cross lines.
3. **Label Thresholds:** Minimum 4 labels for 3M questions; minimum 6 to 8 accurate labels for 5M questions.
4. **Structural Polarity:** Clearly show cis/trans faces (Golgi), 9+2 doublets (axoneme), cristae folds (mitochondria).

Master Board Priority Order:

Tier 1 (Highest 5M/3M Priority): Fluid Mosaic Model • Mitochondrion L.S. • Chloroplast Ultrastructure • Chromosome Centromere Types

Tier 2 (High 3M Probability): Cilia/Flagella 9+2 Axoneme • Prokaryotic Bacterium

Tier 3 (Key Organelles & Comp.): Plant vs Animal Cell • Endomembrane System (ER & Golgi) • Nucleus & Ribosomes